

Wave Release Coordination under Vertical Resource Constraints in Multi-Story AMR Warehouses

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1 Methodology

We address optimization (1) by replacing direct search over the combinatorial decision space $W \subseteq \mathcal{O}$ with two analytical tools that operate on the structured-feature representation $\Phi(W)$ defined in Section 3. The first, a *Bound-and-Gap framework*, decomposes the value of structured-feature information for any fixed partition of Φ space into two non-negative components, yielding a per-regime diagnostic of where wave-design effort pays off. The second, a *Model-Dominance Hedge Rule*, collapses the minimax decision over competing elevator abstractions $\{M_1, M_2\}$ into a one-line closed-form rule under a per-wave dominance condition, with a bound on worst-case loss when dominance is approximate. Both tools take optimization (1) as their starting point and target the per-cell tactical decision: which corner of Φ space to release a wave from.

Bound-and-Gap framework. Fix an operating regime, namely a tuple of system parameters $(F, |\mathcal{A}|, E, c)$ together with an elevator model M . Partition Φ space into a finite set of corners $\mathcal{Q}$ (in our experiments, the four quartile corners on (C, I)). For each corner $q \in \mathcal{Q}$, let $m_q = m_q(M)$ denote the median makespan of waves whose Φ falls in q when simulated under M , and let $m_0 = m_0(M)$ denote the median makespan over a random pool of waves of the same cardinality. The partition’s *value of structured-feature information* is then

$$\text{GAP}(\mathcal{Q}, M) := \frac{m_{q_{\max}} - m_{q_{\Phi}}}{m_0}, \quad (7)$$

where $q_{\max} := \arg \max_{q \in \mathcal{Q}} m_q$ is the worst corner (oracle-pessimistic upper anchor), and q_{Φ} is the corner selected by a Φ -informed policy that picks the empirically best corner on Φ .

Theorem 1 (Bound-and-Gap decomposition). *Under any fixed partition $\mathcal{Q}$ of Φ space and any elevator model M , the value of structured-feature information decomposes as*

$$\text{GAP}(\mathcal{Q}, M) = H_{\text{up}}(\mathcal{Q}, M) + M_{\Phi}(\mathcal{Q}, M), \quad (8)$$

where $H_{\text{up}} := (m_{q_{\max}} - m_0)/m_0$ is the partition-intrinsic upper-tail headroom and $M_{\Phi} := (m_0 - m_{q_{\Phi}})/m_0$ is the Φ -policy slack. Both components are non-negative.

Proof. By definition (7), $\text{GAP} = (m_{q_{\max}} - m_{q_{\Phi}})/m_0$. Adding and subtracting m_0 in the numerator gives

$$\text{GAP} = \frac{m_{q_{\max}} - m_0}{m_0} + \frac{m_0 - m_{q_{\Phi}}}{m_0} = H_{\text{up}} + M_{\Phi}.$$

Non-negativity of H_{up} follows because $m_{q_{\max}} \geq m_0$: the worst corner has median makespan at least as large as the random-pool median, since $m_{q_{\max}}$ is selected by maximization. Non-negativity of M_{Φ} follows because $m_0 \geq m_{q_{\Phi}}$: the Φ -informed policy selects a corner whose median is no worse than the random-pool median in expectation under the partition’s optimal selection rule. $\square$

Corollary 1 (Refinement monotonicity). *The partition-intrinsic upper-tail headroom H_{up} is monotone under partition refinement: if $\mathcal{Q}'$ refines $\mathcal{Q}$, then $H_{\text{up}}(\mathcal{Q}') \geq H_{\text{up}}(\mathcal{Q})$. This holds because refining can only weakly increase $m_{q_{\text{max}}}$ relative to $m_{q_{\text{max}}}$. Corollary 1 explains why finer partitions inflate the upper-bound term mechanically rather than capturing genuinely new information.*

Operational interpretation of Theorem 1. Theorem 1 partitions per-cell wave-design value into two diagnostic axes that operators can act on independently: H_{up} measures how much room there is between the worst corner and the random-pool baseline (a partition-intrinsic upper anchor on what wave structure could possibly buy), while M_{Φ} measures how much slack the Φ -informed policy leaves against the random-pool baseline (a policy-intrinsic gap). Operators read this diagnostic per regime: when H_{up} dominates and M_{Φ} is small, additional elevator capacity is the right lever; when M_{Φ} dominates and H_{up} is small, better Φ -policy design is the right lever; when both are small, neither lever pays off and the wave layer can be left to a heuristic. Section 5 verifies (8) empirically across 18 (regime, size) cells, every one of which satisfies both non-negativity conditions.

Model-Dominance Hedge Rule. When operations managers face uncertainty about which elevator model best represents their warehouse (for example, whether throughput aggregation M_1 or true co-occupancy batching M_2 better captures their site), the natural conservative decision is the *minimax wave-corner selection*

$$c^* := \arg \min_{c \in \mathcal{Q}} \max\{m_c(M_1), m_c(M_2)\}, \quad (9)$$

which picks the corner with the best worst-case median makespan over the two models. Solving (9) directly requires evaluating both models on every corner, which is operationally costly under live deployment. We show that under a mild per-wave dominance condition, the minimax decision admits a one-line closed-form solution.

Theorem 2 (Model-Dominance Hedge Rule). *Suppose for every wave W in the corner pool, $\mathcal{T}_{M_2}(W) \geq \mathcal{T}_{M_1}(W)$ almost surely (per-wave dominance). Then the minimax wave-corner selection (9) collapses to*

$$c^* = \arg \min_{c \in \mathcal{Q}} m_c(M_2). \quad (10)$$

Proof. Per-wave dominance, $\mathcal{T}_{M_2}(W) \geq \mathcal{T}_{M_1}(W)$ for every wave, implies $m_c(M_2) \geq m_c(M_1)$ for every corner c , since the median is monotone under per-wave domination of CDFs. Therefore $\max\{m_c(M_1), m_c(M_2)\} = m_c(M_2)$ for every c , and (9) reduces to $\arg \min_c m_c(M_2)$, the corner preferred by the dominant model M_2 alone. $\square$

Corollary 2 (One-sided ε -bound for relaxed dominance). *Suppose per-wave dominance holds only with probability $1 - \varepsilon$, namely $\Pr[\mathcal{T}_{M_2}(W) \geq \mathcal{T}_{M_1}(W)] \geq 1 - \varepsilon$. Then the corner-wise median bound*

$$m_c(M_1) - m_c(M_2) \leq U_c(\varepsilon) := F_{M_2,c}^{-1}\left(\frac{1}{2} + \varepsilon\right) - F_{M_2,c}^{-1}\left(\frac{1}{2}\right) \quad (11)$$

holds for every c , where $F_{M_2,c}$ is the CDF of $\mathcal{T}_{M_2}(W)$ on corner c . The hedge rule (10) thus retains its argmin choice as long as the inter-corner M_2 -median gaps exceed $\max_c U_c(\varepsilon)$, with worst-case loss against the model-specific optimum bounded by $\max_c U_c(\varepsilon)/m_0$.

Operational interpretation of Theorem 2. Theorem 2 collapses the minimax wave-corner decision over competing elevator models to a one-line closed-form rule, eliminating the need for online model selection. The operator computes $m_c(M_2)$ once for each corner (using the more conservative model M_2 alone), and picks the argmin. Corollary 2 quantifies the rule's

robustness when dominance is only approximate: the operator can compute $U_c(\varepsilon)$ from the M_2 -CDF alone, without needing M_1 , and check whether the inter-corner M_2 -gap exceeds it; if yes, the argmin choice (10) is unchanged. In our experiments (Section 5), per-wave $\mathcal{T}_{M_2} \geq \mathcal{T}_{M_1}$ holds in 92.5–100% of waves across all 15 (regime, arm) cells, well within the regime where Corollary 2’s bound is tight.

Sections 5 and 6 validate these tools through pre-registered simulation experiments. The Bound-and-Gap diagnostic prospectively classifies every (regime, model) cell on whether destination-clustered batching beats first-come-first-served dispatch (6/6 correct), and the Hedge Rule’s underlying dominance condition holds robustly at elevator capacities $c \in \{2, 3, 4, 5\}$ beyond the $c = 2$ case for which the rule was originally derived.